

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Ecological monitoring & new technologies - Set 1

Mark scheme

AQA spec 3.1.2.3.5-3.1.2.3.6; The living environment; textbook pp. 402-407

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>2 marks for each point (partial credit for a partly correct point).</i> • Size: shows whether the population is viable / how urgent action is; sets a baseline to judge success; needed to estimate carrying capacity and sustainable off-take (2) • Distribution: shows which habitats / areas are used and whether the population is fragmented into non-viable isolated groups needing corridors; identifies key areas to protect (2) • Survival rate (mortality): identifies the life stage and causes of death (predation, disease, poaching) so management targets the limiting factor (2) • Age structure: many young indicates a growing population; few young / ageing population indicates breeding failure or high juvenile mortality even if adults are numerous; k-selected species need breeding adults protected (2)	8
01.2	<i>1 mark for each point.</i> • Direct counts from vehicles / aircraft / drones along transects, or camera traps with individual recognition (1) • Sub-sample counts in known areas scaled up to the whole reserve; mark-release-recapture (Lincoln index) for smaller species (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 393-394, 400-402; AQA spec 3.1.2.3.5.	

Question	Marking guidance	Marks
02.1	<i>2 marks for each point (partial credit for a partly correct point).</i> - Tracking: continuous data on movements, territory size, migration routes and habitat use without disturbance, over large areas and long periods; acoustic tags in seawater; geolocators for small birds; more representative than occasional sightings (2) - DNA: identifies individuals, gene pool size / inbreeding and regional populations (tracing ivory, timber, fish); eDNA detects rare or hidden aquatic species (great crested newt, invasive carp) from shed cells without capture (2) - Image recognition: software identifies individuals from unique markings (tiger stripes, whale shark spots, fin damage) in camera-trap / drone / satellite images, giving population size, lifespan, social structure from large image sets quickly and objectively (2) - Acoustic: bat detectors, hydrophones and sonograms record species that are hard to see (bats, cetaceans, insects), giving presence, abundance and activity continuously, day and night (2)	8
02.2	<i>1 mark for each point.</i> - Data collected continuously / over large areas / at night rather than at a few times and places a researcher can visit (1) - Less disturbance so animals behave naturally; larger sample sizes; objective rather than subjective identification (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402-406; AQA spec 3.1.2.3.6.	
03.1	<i>Correct answer only.</i> B - radio waves do not travel through seawater; acoustic or satellite tags transmit position (1)	1
03.2	<i>2 marks for each point (partial credit for a partly correct point).</i> - Shows survival, dispersal and breeding of released individuals so success can be assessed and future releases improved (2) - Radio / GPS tags or collars; leg rings / wing tags on birds; camera traps; DNA from droppings to confirm identity and diet (2)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 393-394, 402-407; AQA spec 3.1.2.3.5-3.1.2.3.6.	
04.1	<i>Correct answer only.</i> Environmental DNA (eDNA) analysis (1)	1
04.2	<i>1 mark for each point.</i> - Transmitters are too heavy for small birds / their batteries would not last for a long-term study (1) - Geocator tags are small and only record time and light levels; latitude and longitude for each day are worked out when the bird is recaptured and the data downloaded (1)	2
04.3	<i>1 mark for each point.</i> - Each separate (regional) population has its own unique gene pool (1) - DNA from seized ivory can be compared with the database to identify where the elephant came from, so poaching hotspots / trade routes can be targeted (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 404-405; AQA spec 3.1.2.3.6.	

Question	Marking guidance	Marks
05.1	<i>1 mark for each point.</i> • Images / maps of large areas; land-use change, deforestation, flooding (1) • Vegetation density and surface temperature (1) • Wind velocity, sea-surface altitude, wave height, oil pollution (1)	3
05.2	<i>1 mark.</i> • Microwaves pass through clouds (unlike visible light) (1)	1
05.3	<i>1 mark for each point.</i> • Very large / remote areas can be surveyed quickly and repeatedly so changes over time are detected (1) • No need for researchers to visit inaccessible areas, reducing cost and disturbance; data are standardised across the whole area (1)	2
05.4	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> • Motion-sensitive cameras placed along trails record animals passing, using visible light by day and infrared at night (1) • Each tiger has a unique pattern of facial / body stripes (1) • Images are compared with the database so individuals are recognised rather than counted twice (1) • The number of different individuals recorded gives a minimum population; repeated captures give territory size, movements, lifespans and social groupings (1) • Limitation: only works for species with unique visible features (1)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402-403, 406-407; AQA spec 3.1.2.3.6.	
06.1	<i>1 mark for each point.</i> • Its reading is adjusted to match a meter known to be accurate so that the results are accurate / comparable (1) • If it cannot be adjusted the percentage error should be recorded so the results can be corrected (1)	2
06.2	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Droppings: diet, gender, territory size (1) • Owl pellets: diet (1) • Feeding marks (eg chewed hazel nuts): presence of species such as dormice (1) • Nests / burrows: presence and breeding (1) • Tracks / footprints: presence, movements (1) • Territorial marks (scratching posts): presence and territory (1)	3
06.3	<i>1 mark for each point.</i> • Bat detectors / sonograms record the ultrasonic sounds bats produce, which differ between species (1) • The recordings provide evidence of presence, abundance and activity of species that cannot easily be seen (1)	2
06.4	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Marked individuals mix freely with the rest of the population (1) • Marks do not fall off or wear away, and marking does not affect survival (1) • The population does not change during the study (no births, deaths or migration) (1) • All individuals have an equal chance of being caught / animals are mobile and can be caught (1)	3

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402, 405, 407; AQA spec 3.1.2.3.5-3.1.2.3.6.	
07.1	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • $(15 - 6) / 6 \times 100$ (1) • = 150% (1)	2
07.2	<i>1 mark for each point.</i> • Cells shed by the newts into the water contain their DNA (1) • Water samples are analysed for DNA sequences unique to the species (1)	2
07.3	<i>1 mark for each point.</i> • Catches depend on trapping effort, weather and newt activity on the night, so they vary from year to year (1) • Trapping only samples one pond and one part of the population, whereas eDNA surveys all 20 ponds with a standard method (1)	2
07.4	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Image databases of unique markings (tiger stripes, whale shark spots, dolphin fins) giving territory size, movements, lifespan and social groups; limited to species with unique features • Motion-sensitive and infrared cameras, CCTV of nests with less disturbance; marking (rings, wing tags) • DNA: eDNA to detect presence (great crested newts, invasive carp); DNA databases to identify regional populations and trace timber, ivory and fish • Auditory monitoring: bat detectors and sonograms for presence, abundance and activity • Position monitoring: radio, GPS and acoustic transmitters; geolocator tags for small birds; data recorders; carrier systems - ROVs / drones, AUVs, balloons, aircraft, satellites, animals • Satellites: visible light for land use and deforestation, infrared for vegetation and temperature, radar for wind, waves, sea level and oil; GPS for tracking • Evaluation: greater coverage, less disturbance and continuous data vs cost, battery / weight limits, calibration, need for ground-truthing, ethical issues of tagging; indirect evidence and traditional methods still needed • Justified conclusion	9
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402-407; AQA spec 3.1.2.3.5-3.1.2.3.6.	